

Fait à la maison avec feuille exercices donnée

#5) $HR = \frac{P_{\text{eau}}}{P_{\text{eau}}^0} = \frac{1,228 \cdot 10^3 \text{ Pa}}{2715,71 \text{ Pa}} = 45,21\%$

$P_{\text{eau}}^0 \text{ à } 22,5^\circ\text{C} = 10^{\frac{1025 - 1750}{22,5 + 235}} = 2715,71 \text{ Pa}$

$P_{\text{eau}} \text{ à } 10^\circ\text{C} = 1,228 \cdot 10^3 \text{ Pa}$

#12



$T=10 \rightarrow T=80^\circ\text{C} \quad 8,314$

$P \cdot V = n \cdot R \cdot T$

$V_{\text{eau}} = \frac{20\text{L}}{1000\text{L} \rightarrow 1\text{m}^3} = 0,02 \text{ m}^3$

$m_{\text{eau}} = 0,02 \text{ m}^3 \times 1000 \text{ Kg/m}^3 = 20 \text{ Kg}$

$m_{\text{eau évaporé}} = 0,10 \times 20 \text{ Kg} = 2 \text{ Kg}$

$n_{\text{vapeur eau}} = \frac{2000\text{g}}{18,015/\text{mol}} = 111,05 \text{ mol}$

$T(\text{K}) = 80^\circ\text{C} + 273,15 = 353,15 \text{ K}$

$P_{\text{eau } 80^\circ\text{C}} = 47,39 \text{ kPa} = 47,39 \cdot 10^3 \text{ Pa}$

$V = \frac{111,05 \cdot 8,314 \cdot 353,15}{47,39 \cdot 10^3} = 6,88 \text{ m}^3$



$\text{masse } \text{CaCO}_3 = \frac{95}{100} \times 200 \text{ Kg} = 190 \text{ Kg}$

$M_{\text{Ca}} = 40,08 \text{ g/mol}$

$M_{\text{C}} = 12,01 \text{ g/mol}$

$M_{\text{O}} = 16 \text{ g/mol}$

$M_{\text{CaCO}_3} = 40,08 + 12,01 + 3 \cdot 16 = 100,09 \text{ g/mol}$

$M_{\text{CaO}} = 40,08 + 16 = 56,08 \text{ g/mol}$

$n_{\text{CaCO}_3} = \frac{190 \cdot 10^3 \text{ g}}{100,09 \text{ g/mol}} = 1898 \text{ mol}$

$n_{\text{CaO}} = n_{\text{CaCO}_3}$

$m_{\text{CaO}} = 1898 \times 56,08 = 107\,958 \text{ g} = 107,958 \text{ Kg}$

Ex #27



4g de NaHCO_3

0,0800 mol H^+

$$M_{\text{NaHCO}_3} = 22.99 + 1.008 + 12.01 + 3 \cdot 16 = 84.008 \text{ g/mol}$$

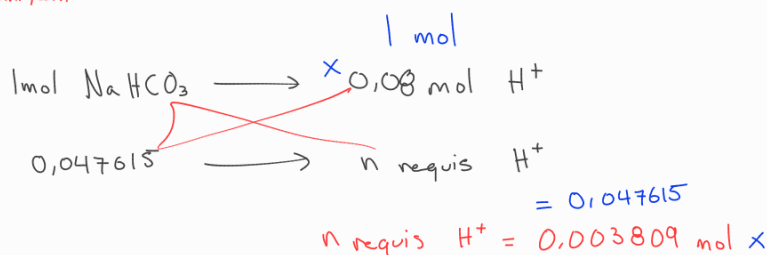
$$n_{\text{NaHCO}_3} = \frac{m}{M} = \frac{4}{84.008} = 0.047615 \text{ mol}$$

$$\frac{n_{\text{NaHCO}_3}}{\text{Coeff stochio}} = \frac{0.047615}{1} = 0.047615$$

$$\frac{n_{\text{H}^+}}{\text{Coeff stochio}} = \frac{0.0800}{1} = 0.0800$$

$$0.047615 < 0.0800$$

$\text{NaHCO}_3 \rightarrow$ Réactif limitant



b) $\% \text{ excès} = \frac{\text{Qty fournie} - \text{Qty requis}}{\text{Qty fournie}} = \frac{0.0800 - 0.003809 \times}{0.0800} \times 100 = 95.24\% \times$
40.48%

c) conversion limitant 90%
rendement 85%

$V_{\text{CO}_2} = ?$, $P = 101.3 \text{ kPa}$, $T = 150^\circ\text{C}$

$$P \cdot V = n \cdot R \cdot T$$

$R = 8.314$

$$n_{\text{NaHCO}_3 \text{ réagis}} = 0.047615 \times 0.90 = 0.042854 \text{ mol}$$



$$n_{\text{CO}_2} = 0.042854$$

$$V = \frac{0.042854 \cdot 8.314 \cdot (150 + 273.15)}{101.3 \cdot 10^3} = 0.001488 \text{ m}^3 \times 10^3 = 1.48829 \text{ L}$$

0,0

$$1000 \text{ L} = 1 \text{ m}^3$$

